

electron/git-service.cjs

Kind: added · Baseline: 8e0dc031dcfc · Target: NovaDiff · Generated: 2026-05-19T21:13:03.437Z

Overview

`electron/git-service.cjs` (lines 1-229) replaces scattered shell calls with a single module that centralises all Git interactions for the Electron renderer.

Key changes

- **Imports added** (lines 3-6): `const fssync = require("node:fs"); const path = require("node:path"); const os = require("node:os"); const { spawnSync } = require("node:child_process");`
- `gitExecutable()` (lines 8-10) returns `"git.exe"` on Windows, otherwise `"git"`.
- `runGit()` / `tryRunGit()` (lines 12-27, 30-36) wrap `spawnSync` with a 120s timeout, error handling, and normalise output.
- **Repository checks:** `isGitRepo()` (lines 38-45) verifies a Git repo.
- **Status parsing:** `parsePorcelainStatus()` (lines 47-74) and `getRepoStatus()` (lines 75-123) expose branch, upstream, dirty files, remotes, etc.
- **URL parsing:** `parseGithubSlugFromUrl()` (lines 124-136) extracts owner/repo from GitHub URLs.
- **Git actions:** `stageAll()` (137-140), `commit()` (142-146), `push()` (148-154).
- **Worktree helpers:** `ensureWorktreeBase()` (156-160), `removeDirSafe()` (162-168), `checkoutRefWorktree()` (174-182), `cleanupNovadiffWorktrees()` (184-186).
- **PR comparison:** `preparePrCompareRoots()` (191-205) creates detached worktrees for base/head comparison.
- **Tooling detection:** `detectGitTooling()` (207-214) reports Git availability and version.
- All functions are exported via `module.exports` (lines 216-229).

Impact

- **Maintainability:** Single source of truth for Git logic reduces duplication.
- **Cross-platform:** `gitExecutable()` and `spawnSync` provide consistent behavior on Windows, macOS, and Linux.
- **Observability:** Errors from `runGit()` include the underlying message, aiding debugging.
- **Feature completeness:** Worktree support enables accurate PR diffing without altering the main repo.

Risks & follow-ups

- **UI blocking:** `spawnSync` is synchronous; impact on large repos is unknown from the diff.
- **Worktree cleanup:** `removeDirSafe()` may race if multiple instances run concurrently; behavior not covered by the diff.

- **CI environments:** `detectGitTooling()` should be verified in containers lacking Git; failure handling is present but not tested in the diff.
- **Git version compatibility:** `parsePorcelainStatus()` assumes the current porcelain format; compatibility with older Git releases is unknown from the available evidence.